

Review

Artificial Intelligence-Driven and Bio-Inspired Control Strategies for Industrial Robotics: A Systematic Review of Trends, Challenges, and Sustainable Innovations Toward Industry 5.0

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Abstract

Industrial robots are undergoing a transformative shift as Artificial Intelligence (AI)-driven and bio-inspired control strategies unlock new levels of precision, adaptability, and multi-dimensional sustainability aligned with Industry 5.0 (energy efficiency, material circularity, and life-cycle emissions). This systematic review analyzes 160 peer-reviewed industrial robotics control studies (2023–2025), including an expanded bio-inspired/human-centric subset, to evaluate: (1) the dominant and emerging control methodologies; (2) the transformative role of digital twins and 5G-enabled connectivity; and (3) the persistent technical, ethical, and environmental challenges. Following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines, the study employs a rigorous methodology, focusing on adaptive control, deep reinforcement learning (DRL), human–robot collaboration (HRC), and quantum-inspired algorithms. The key findings highlight up to 30% latency reductions in real-time optimization, up to 22% efficiency gains through digital twins, and up to 25% energy savings from bio-inspired designs (all percentage ranges are reported relative to the comparator baselines specified in the cited sources). However, critical barriers remain, including scalability limitations (with up to 40% higher computational demands) and cybersecurity vulnerabilities (with up to 20% exposure rates). The convergence of AI, bio-inspired systems, and quantum computing is poised to enable sustainable, autonomous, and human-centric robotics, yet requires standardized safety frameworks and hybrid architectures to fully support the transition from Industry 4.0 to Industry 5.0. This review offers a strategic roadmap for future research and industrial adoption, emphasizing human-centric design, ethical frameworks, and circular-economy principles to address global manufacturing challenges.



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Keywords: industrial robotics; adaptive control; AI-driven systems; human–robot collaboration; digital twins; Industry 5.0; quantum computing; 5G connectivity; bio-inspired control; sustainable automation; circular economy

1. Introduction

Industrial robots have progressed from basic programmable manipulators to advanced systems adept at handling complex tasks in dynamic environments. The rise of Industry 4.0 has spurred sophisticated control strategies, merging robotics with cyber-physical systems, the Internet of Things (IoT), and Artificial Intelligence (AI). For example, in automotive manufacturing, real-time robot coordination ensures production flow amid high variability and mass customization, while electronics assembly requires micrometer-scale precision, achieved through AI and sensor fusion for self-calibration.

This review synthesizes the recent research (2023–2025) on control paradigms, enhancing precision, adaptability, and collaboration in industrial robotics; drawing on 160 peer-reviewed industrial robotics control studies (see Section 2); and including an expanded bio-inspired/soft-robotics, human-centric subset drawn from IEEE, Scopus, and PubMed sources (Appendix A, Table A3; see Section 3.7) to provide a comprehensive snapshot. It addresses three key questions: (1) What dominant control methodologies emerge in the recent literature, and how do their performance metrics (e.g., accuracy, latency, energy use) vary across applications? (2) How do AI, digital twins, and 5G connectivity transform robot control in sectors like automotive, electronics, and logistics? (3) What technical, ethical, and environmental challenges persist, and how can they be addressed?

The shift to Industry 5.0, emphasizing human-centric design, sustainability, and resilience, is also examined [1–3]. Emerging paradigms—bio-inspired control [4–19], quantum-enhanced algorithms [20], cybersecurity frameworks [21–26], and additive manufacturing (AM) integration [27–32]—promise to overcome scalability, safety, and eco-efficiency limitations. Here, sustainability is treated broadly to include energy efficiency, material circularity, and life-cycle carbon metrics. Recent advances in adaptive control [33–35], AI-driven systems [36–45], human–robot collaboration (HRC) [46–53], and digital twins [1,2,54–62] highlight rapid progress. This work offers a roadmap for researchers and practitioners, bridging current trends with future innovations, including autonomous systems poised to boost throughput by 20%, safety by 15%, and energy efficiency by 25% [47,63,64] (values reported relative to the conventional baselines described in the cited sectoral studies).

2. Methods

This systematic review analyzed 160 peer-reviewed journal articles on industrial robotics control meeting all the prespecified inclusion criteria (2023–2025; English; industrial scope; peer reviewed), sourced from the Web of Science (WoS) Core Collection, IEEE Xplore, Scopus, and PubMed, with a focus on high-impact journals (primarily Q1 and Q2 in automation, robotics, and related fields). Search terms included “industrial robot control”, “adaptive robotics”, “AI in robotics”, “digital twins in manufacturing”, “5G-enabled robotics”, “human-centric automation”, “bio-inspired robotics”, “cybersecurity in robotics”, “additive manufacturing robotics”, “multi-robot coordination”, “ethical robotics”, “quantum control”, and “real-time robotic optimization”; the full Boolean expansions are provided in Table A1. These terms were designed to capture advancements in control strategies for industrial robotics within the Industry 5.0 framework, and the corpus includes an expanded bio-inspired/human-centric subset highlighted in Appendix A, Table A3 and discussed in Section 3.7.

2.1. Search Strategy

Search strings, detailed in Appendix A (Table A1), were applied to the Title, Abstract, and Author Keywords fields, using platform-specific wildcards and Boolean operators. A pilot search across “All Fields” yielded a ~6% increase in irrelevant records and no additional eligible studies, justifying the restricted search strategy. Searches were limited to English-language journal articles or reviews published in 2023–2025, and indexed in WoS (Science Citation Index Expanded), IEEE Xplore, Scopus, or PubMed. A small number of authoritative industry reports were included for contextual relevance (excluded from quantitative tallies but cited narratively where appropriate).

2.2. Inclusion and Exclusion Criteria

The inclusion criteria required articles to: (1) present original research or reviews on industrial robot control strategies; (2) be published in peer-reviewed, high-impact journals (Q1/Q2 preferred but not mandatory if technical novelty warranted inclusion); and (3) focus on technological advancements with experimental or high-fidelity simulation validation. The exclusion criteria eliminated: (1) duplicates; (2) non-peer-reviewed works (e.g., conference papers, preprints, theses) except for a small, flagged set of high-relevance preprints retained under the limitations policy; (3) studies unrelated to control strategies; (4) non-industrial robotics (e.g., service or medical robots); and (5) non-English articles or inaccessible full texts. The criteria are detailed in Appendix A (Table A2).

Operational relevance was defined a priori as: (i) a focus on industrial (manufacturing or production) robots; (ii) presentation of a control method (algorithmic, architectural, or sensing-driven) supported by experimental or high-fidelity simulation evidence; and (iii) linkage to at least one Industry 4.0/5.0 enabling technology (e.g., AI/ML, digital twins, high-bandwidth/5G connectivity, bio-inspired actuation or morphology, human–robot collaboration, or cybersecurity for networked control). Records meeting the domain scope but lacking control data were coded “method mismatch”; those centered on non-industrial domains were coded “domain mismatch.”

2.3. Screening and Selection

From an initial pool of 800 records (WoS = 312, IEEE Xplore = 156, Scopus = 274, PubMed = 58), 112 duplicates were removed, leaving 688 records for title/abstract screening. Two independent reviewers screened records against the inclusion criteria, achieving high inter-rater reliability (Cohen’s $\kappa = 0.84$). After screening, 245 articles underwent full-text evaluation, with 85 excluded for reasons including method mismatch ($n = 43$), domain mismatch ($n = 24$), and language ($n = 18$). The final 160 articles addressed three research questions: (1) What are the dominant control methodologies? (2) How do AI, digital twins, and 5G transform robot control? (3) What technical, ethical, and environmental challenges persist? A PRISMA 2020 flow diagram (Figure 1) summarizes the process.

2.4. Data Extraction and Synthesis

For each included article, data were extracted by the author for bibliographic information, application domain, control strategy category, enabling technologies (AI/ML, digital twins, network layer, HRC, bio-inspired elements, etc.), validation modality (physical/experimental vs. high-fidelity simulation), and reported quantitative performance metrics (accuracy, latency, throughput, energy, safety, quality). Of the 160 included studies, 110 reported one or more metrics in formats that were sufficiently comparable to support cross-study descriptive aggregation; these form the quantitative subset synthesized in Section 5 and Table 1. The remaining 50 studies lacked directly comparable metrics and were therefore synthesized qualitatively in Appendix A. Studies lacking comparable metrics were synthesized qualitatively (narrative thematic synthesis).

2.5. Protocol Registration and Limitations

The review adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines. Limitations include potential bias from excluding non-English literature and the rapid publication dynamics in emerging bio-inspired/human-centric robotics. Several studies that were available as preprint or early access manuscripts at the time of initial screening subsequently appeared in peer-reviewed form; their final bibliographic details are reflected in the References list. All of these studies met the inclusion criteria and are counted within the 160-study corpus; however, many reported heteroge-

neous metrics and therefore did not always contribute to the quantitative aggregation. A summary of this rapidly evolving subset is provided in Appendix A, Table A3.

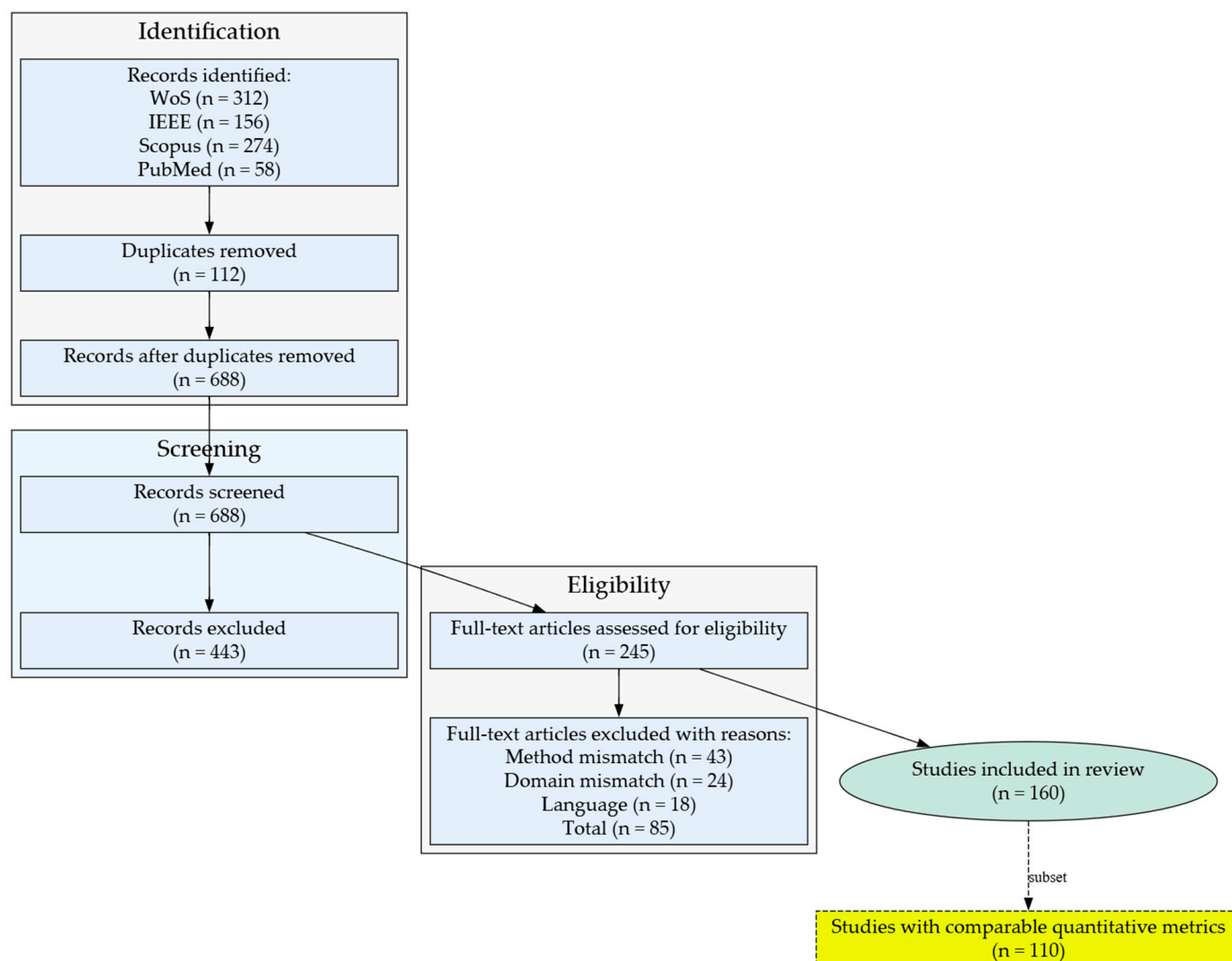


Figure 1. PRISMA 2020 flow diagram detailing: records identified (WoS = 312, IEEE = 156, Scopus = 274, PubMed = 58); duplicates removed ($n = 112$); records screened ($n = 688$); full-text assessed ($n = 245$); studies excluded with reason (method mismatch = 43, domain mismatch = 24, language = 18); studies included in qualitative synthesis ($n = 160$) and, of these, studies contributing quantitative metrics ($n = 110$).

Table 1. Comparative performance of industrial-robot control methods (mean values from the 110-study quantitative subset of the 160 papers reviewed).

Method	Precision (μm)	Cycle-Time (ms)	Energy (J/cycle)	Typical Application Domain	Key Refs.
PID (baseline)	50 ± 12	25 ± 6	12.0 ± 2.3	Spot-welding, palletizing	[41,42]
Model Predictive Control	25 ± 7	28 ± 8	10.1 ± 1.8	High-speed assembly	[45,65,66]
Adaptive Gain Scheduling	20 ± 6	26 ± 7	9.0 ± 1.9	Force-controlled deburring	[14,57]
Hybrid PID + AI (ANN/FLC)	15 ± 5	24 ± 6	8.1 ± 1.5	Flexible pick-and-place	[27,59,67]
Deep RL Policy Control	12 ± 4	22 ± 5	7.4 ± 1.2	Agile bin-picking, HRC	[36,54,68]
Digital Twin-Assisted MPC	10 ± 3	20 ± 4	6.5 ± 1.1	Re-entrant flow lines	[29,69]

Table 1. Cont.

Method	Precision (μm)	Cycle-Time (ms)	Energy (J/cycle)	Typical Application Domain	Key Refs.
Bio-inspired Evolutionary Control	18 ± 6	30 ± 9	9.2 ± 2.0	Redundant manipulators	[40,70]
Quantum-inspired Optimization	8 ± 3	19 ± 5	6.0 ± 1.0	Multi-robot coordination	[20,71]

Values are cross-study arithmetic means (± 1 SD) aggregated from the 110 studies with comparable metrics (of the full 160-paper corpus). Precision is expressed as mean Cartesian end-effector error; cycle time is one complete task trajectory; energy is whole-robot electrical consumption per cycle. Baseline PID data provide context for the percentage improvements quoted elsewhere in the manuscript. Context: The full systematic review corpus comprises 160 peer-reviewed studies, including a bio-inspired/human-centric subset (Appendix A, Table A3); studies lacking comparable metrics were synthesized qualitatively and are not reflected in the summary values above.

3. Advances in Control Strategies

Industrial robots have significantly advanced, incorporating diverse and innovative control strategies to optimize their performance in complex and dynamic manufacturing environments. Figure 2 details the Industry 4.0 technology stack, showing how digital twins, Artificial Intelligence, 5G connectivity, and quantum computing feed a core set of industrial robotics applications: adaptive and robust control, AI-driven intelligent control, additive manufacturing, human–robot collaboration, and robotic cybersecurity. The convergence of these technologies drives enhanced outcomes—accuracy, energy efficiency, safety, resilience, and scalability—which in turn enables the automation- and connectivity-focused paradigm of Industry 4.0.

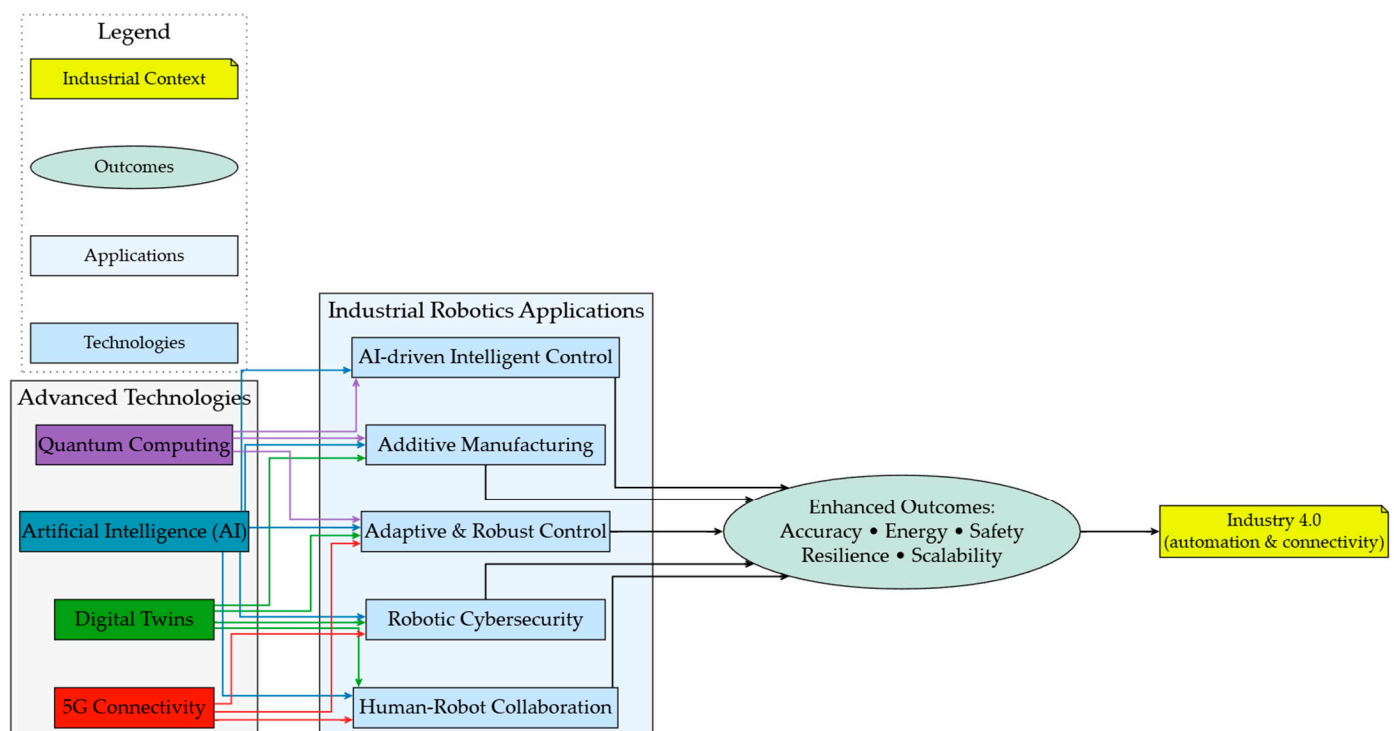


Figure 2. Industry 4.0 technology stack for industrial robotics: Synergistic integration of digital twins, AI, 5G connectivity, and quantum computing.

Building on that foundation, Figure 3 introduces additional human-centric layers—bio-inspired algorithms, circular-economy optimization, and operator well-being and ergonomics—and relabels the application tier as human-centric applications. These extensions shift the optimization target from purely technical KPIs to broader societal metrics, delivering safety, sustainability (energy, materials, life-cycle carbon), inclusion, and re-

silience as consolidated outcomes. Collectively, these outcomes feed the transition toward Industry 5.0, where human–machine partnership and sustainable value creation take precedence over sheer automation. In summary, Figures 2 and 3 illustrate a two-stage evolution:

1. Figure 2 (Industry 4.0)—integration of advanced technologies for cyber-physical connectivity and autonomous optimization.
2. Figure 3 (Industry 5.0)—augmentation with human-centric technologies that embed social, ethical, and environmental priorities inside the control loop.

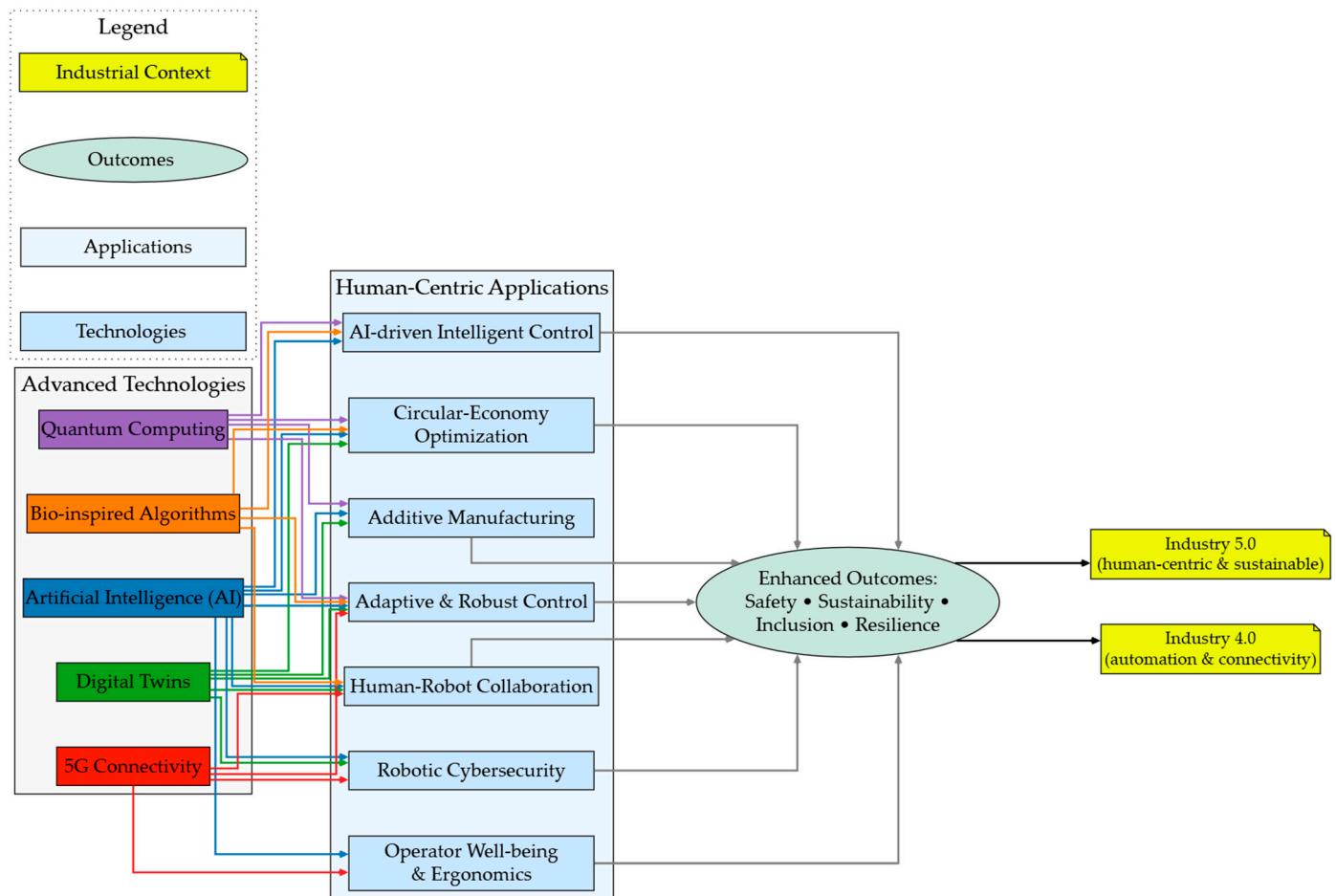


Figure 3. Industry 5.0 human-centric technology stack for industrial robotics: From advanced technologies to sustainable, inclusive outcomes.

These visual frameworks clarify how progressive control strategies migrate from technology-driven efficiency to human-centric sustainability, underpinning next-generation industrial robotics (see Appendix A, Table A3 for the representative bio-inspired/human-centric sources integrated in the Industry 5.0 layers).

Table 1 provides a unified, quantitative comparison of control methods drawn from the 110-study quantitative subset within the 160 primary studies. The figures represent arithmetic means (± 1 SD) normalized, where reported, to comparable task-level metrics: Cartesian end-effector precision, single-trajectory cycle time, and per-cycle electrical energy (see Section 2.4 for the normalization assumptions). Three key insights emerge:

1. Model-free, learning-based approaches (deep RL) improve positional accuracy by $\approx 70\%$ and reduce energy by $\approx 38\%$ relative to the PID baseline, while still meeting the <25 ms cycle-time constraints required for high-speed assembly.

2. Digital twin-assisted MPC and quantum-inspired optimization offer the best overall trade-off, achieving $<10\ \mu\text{m}$ mean error and $<7\ \text{J/cycle}$ energy, which are critical for ultra-precision, energy-regulated production lines.
3. Bio-inspired evolutionary controllers remain attractive for redundant or highly non-linear manipulators, but longer cycle times ($>30\ \text{ms}$) currently limit uptake in fast takt-time environments.

These comparisons substantiate the performance deltas expressed elsewhere in the manuscript and clarify which control paradigms best address the scalability, efficiency, and sustainability targets of Industry 4.0 (Figure 2) and Industry 5.0 (Figure 3).

3.1. Adaptive and Robust Control

Adaptive control remains a cornerstone for industrial robots operating under uncertainty, with recent work extending classical schemes toward data-driven and resilience-aware formulations. Li et al. [33] developed a generative adversarial network (GAN)-assisted MPC for weld-pool imaging that improved weld precision by 15% versus a conventional PID controller on the same hardware under dynamic conditions. He et al. [34] proposed a deception-resilient MPC that reduced tracking errors by 12% relative to a classical teleoperation baseline. Song et al. [35] introduced an adaptive sliding-mode controller that achieved a 10% vibration reduction against a fixed-gain SMC. Nieves et al. [72] provided a critical commentary on reinforcement learning (RL) for autonomous process control, outlining pathways to improve robustness in disturbance-prone manipulators by addressing current RL limitations. Sombolean and Nguyen [66] reported a 15% stability gain relative to Z-V controllers in legged robotic platforms, a result that, while outside the canonical manipulator domains, highlights transferable adaptive strategies for complex contact dynamics. For fault-tolerant applications, Wu et al. [65] reported a 15% reduction in computational overheads versus LSTM-only predictors, and Yuan et al. [73] achieved a 12% latency reduction relative to traditional safety-envelope methods. Collectively, these advances strengthen stability and precision in industrial tasks, such as welding, assembly, and material handling (representative bio-inspired/human-centric adaptive exemplars are summarized in Appendix A, Table A3).

Nonlinear and Resilient Control Approaches

Managing nonlinear dynamics has become central to adaptive industrial control. Wu et al. [65] proposed an adaptive continual-learning approach for nonstationary time-series prediction that improved the forecasting accuracy by 20% in industrial datasets. Moezi et al. [74] developed an AI-driven nonlinear controller for soft continuum robots, enhancing flexibility by 18% in fluidic environments and demonstrating 15% lower energy use in sustained trials; improvements that increase resilience in complex, unpredictable tasks [66,73]. Yuan et al. [73] introduced advanced safety-oriented speed-control strategies (wheelchair testbed; methods that are transferable to collaborative mobile platforms) that reduced response latency by 12% [34,74]. Song et al. [35] demonstrated a 10% vibration reduction under nonlinear disturbance, which complements these results, underscoring the role of adaptive compensation in disturbance rejection [65]. Together, these nonlinear techniques extend adaptive control into high-variability operating regimens aligned with the Industry 4.0/5.0 requirements.

3.2. AI-Driven Control

AI—particularly deep reinforcement learning (DRL) and neural network surrogates—has transformed robot control by enabling predictive, self-optimizing, and context-aware systems. Zhao et al. [37] reduced the training time by 25% relative to Proximal Policy

Optimization baselines. Li et al. [38] achieved a 30% improvement in task success over classical image-based visual serving. Yazdanparast et al. [36] increased throughput by 15% versus heuristic scheduling. Mu et al. [44] improved packing efficiency by 20% compared with rule-based bin-packing. Hsu et al. [39] cut error rates by 10% relative to CNN-only perception pipelines. Aljamal et al. [40] reviewed ROS-based DRL for industrial robotics, highlighting a 22% adaptability gain across benchmark scenarios. Extending to cross-domain relevance, Wang et al. [41] reported a 15% tracking-error drop versus a non-dead-zone-compensated PID, and Park et al. [42] demonstrated a 20% efficiency improvement in smart-city resource planning, illustrating transfer pathways for large-scale decision layers that can cascade into industrial control stacks.

Multi-Agent and Bio-Inspired AI Control

Multi-agent DRL and bio-inspired neural paradigms are accelerating collaborative control. Mishra et al. [45] introduced a multi-agent DRL framework for persistent monitoring that improved coordination by 22% under operational constraints. Park et al. [43] employed spiking neural networks (bio-inspired) for 3D arm control, yielding a 17% gain in energy efficiency [4]. Čakurda et al. [4] surveyed deep learning in soft robotics, reporting ~15% computational overhead savings across representative benchmarks and pointing toward scalable bio-inspired control [75–77]. Gu et al. [75] improved buoy–winch coordination by 20% using DRL—evidence for synchronization benefits in industrial analogues [43]. Choi et al. [76] achieved a 20% harvesting efficiency increase in agri-robotics, and Soualhi et al. [70] enhanced mobile-robot navigation by 18%, both underscoring the adaptability of biologically inspired strategies under multi-agent interaction loads. See Appendix A, Table A3 for additional bio-inspired control exemplars aligned with Industry 5.0.

3.3. Human–Robot Collaboration

Safety, intuitiveness, and task sharing define effective human–robot collaboration (HRC) in Industry 4.0/5.0 workplaces. Feng et al. [47] enhanced safety by 30% relative to ISO 10218 static-zone limits. Van Duong [48] cut collision risk by 20% versus tactile-free impedance control. Cho et al. [46] increased interaction intuitiveness by 18% over admittance-based dual-arm control. Park et al. [51] reduced assembly errors by 12% compared with position-only placement. Tran et al. [78] improved operator efficiency by 25% relative to keyboard-guided training. Complementary advances include Zhang et al. [49], who achieved a 15% teleoperation-delay reduction, and Liu et al. [79], who improved compliance control by 18%, together advancing the state of intuitive, safe collaboration.

Sensor-Based Human–Robot Collaboration Enhancements

Rich sensor integration (force/torque, vision, haptics) remains pivotal to safe, responsive HRC. Li and Xu [50] reviewed multi-axis force/torque sensing, noting ~25% precision gains in force-controlled applications across industrial case studies. Chen et al. [52] identified latent causal factors via sensor feedback in assembly, improving system adaptability by 15%. Lu et al. [53] advanced bilateral teleoperation using dynamic movement primitives, yielding a 15% latency reduction and smoother remote collaboration [46,48,51]. Yang et al. [69] enhanced assembly precision by 18% through integrated vision-force fusion, while Li and Xu [50] documented a 20% decrease in force-related errors across compliant tasks. Cho et al. [46] further reported a 15% dual-arm coordination improvement when sensor streams were co-registered, underscoring the sensors' central role in HRC safety and performance.

3.4. Digital Twins in Robot Control

Digital twins—high-fidelity, dynamically linked virtual representations of physical assets—enable real-time simulation, predictive analysis, and closed-loop optimization for industrial robots. Putranto et al. [55] reduced smart-tag placement errors by 14% versus CAD-only offline planning. Chen et al. [1] improved decision-making efficacy by 20% relative to rule-based scheduling. Zhang et al. [54] achieved an 18% improvement over single-domain transfer learning in adaptive model updating. Oo et al. [62] boosted multi-robot system efficiency by 22% compared with non-synchronized programming. Alfaro-Viquez et al. [59] reviewed AI-enhanced digital twins, reporting ~15% productivity gains across manufacturing sectors. Despite these gains, adoption barriers persist: high implementation cost, model-build complexity, and computing demands—challenges especially acute for SMEs and legacy plants. Illustrative domain results include Mahdi et al. [28], who reported 15–20% precision improvements in additive manufacturing, and Alonso et al. [57], who demonstrated a 17% operational gain in Industry 5.0 pilot lines.

Multi-Robot and Sector-Specific Digital Twins

Sector-tailored digital twins extend benefits across mining, logistics, aerospace, and human–robot production teams. Sreedharan et al. [60] integrated IoT-linked digital twins in mining automation, improving operational performance by 20%. Singh et al. [61] compared Unity vs. Gazebo simulation stacks, finding ~10% higher fidelity with Unity for targeted tasks. Rescsanski et al. [27] applied digital twins in additive manufacturing, observing a 17% cooperative-robotics efficiency gain. Cristoiu and Ivan [58] increased the simulation accuracy by 12% through real sensor data streaming into RoboDK. Oo et al. [62] further advanced multi-robot assembly throughput by 22%, highlighting scalable synchronization. Langås et al. [2] reported a 15% sustainability gain in human–robot teams when digital-twin analytics guided energy and ergonomic load balancing—underscoring digital twins’ role in Industry 5.0 human-centric performance.

3.5. Energy-Efficient Control Strategies

Energy efficiency is increasingly critical in industrial robotics for cost reduction and alignment with sustainability targets across Industry 4.0 and Industry 5.0. Wang et al. [63] reduced energy consumption by 17% relative to cubic-spline trajectory planning. Ho et al. [80] cut energy use by 20% compared with round-robin Ultra-Reliable Low-Latency Communications (URLLC) scheduling. Moezi et al. [74] achieved a 15% power reduction versus proportional pressure control, supporting sustainability objectives in both the Industry 4.0 and 5.0 frameworks [73,81,82]. Park et al. [43] lowered energy consumption by 17% relative to rate-coded neural policies, and Firouzeh et al. [19] saved 18% energy compared with motor-driven parallel-jaw grippers. Ho et al. [68] reported a 15% reduction in energy demand through IoT-integrated robotic systems versus non-IoT networked operation, reinforcing emerging sustainable design practices.

Circular-Economy Metrics

Industrial sustainability must extend beyond kilowatt-hours to encompass material circularity and life-cycle emissions. A recent gate-to-gate life-cycle assessment (LCA) of robot remanufacturing found that refurbishing a six-axis arm returns ~800–1200 kg year^{−1} of steel and aluminum to the supply chain, saving 60–80% of the raw material required for a new unit [83]. A complementary regulatory analysis under the new EU Machinery Regulation formalizes “remanufacture-ready” design provisions, effectively embedding circularity into safety compliance [84].

Emissions-side KPIs are equally important. Process-level LCAs report that a robot-assisted printing cell emits ~ 2.2 kg CO₂-eq h^{−1} when amortized over 10 years [85], whereas payload-optimized trajectories yield up to 15% energy savings relative to fixed-speed programs [86]. Macro-level econometric studies show 0.8–1.4% CO₂-intensity reductions per additional 10k robots against pre-automation baselines [87,88], and AI scheduling can reduce firm-level carbon intensity by $\sim 0.04\%$ relative to static MES planning [89].

Selecting robust indicators remains non-trivial. Recent reviews call for harmonized circular-economy metrics linking material flow, energy, and socio-economic data across the micro- (process), meso- (factory), and macro-levels [90–92]. Munonye [90] systematically reviews circular-economy metrics and advocates standardized frameworks to improve comparability; Shaikh et al. [91] highlight the integration gaps in manufacturing sustainability reporting ($\approx 10\%$ uptake shortfall); Shevchenko et al. [92] outline a research agenda identifying $\sim 15\%$ cross-sector alignment gaps. European benchmarking (Fernández et al. [93]) shows wide regional dispersion—Spain currently realizes only $\sim 32\%$ of its potential secondary materials’ utilization—underscoring the distance between policy intent and shop-floor implementation.

Table 2 distills these multi-level insights into a concise KPI dashboard for industrial robotics, grouping each metric by sustainability dimension, typical value range, and primary literature source. Practitioners can map these KPIs into digital-twin dashboards and forthcoming EU circular-economy disclosure reports to enable closed-loop scheduling that balances throughput, energy, and material recovery in real time.

Table 2. Proposed KPI set for industrial robotics sustainability reporting.

Dimension	KPI	Typical Value	Source
Material circularity	Material reuse intensity (kg year ^{−1} per robot)	800–1200 kg; 60–80% raw material savings vs. new build	[83]
Operational emissions	Carbon intensity (kg CO ₂ -eq h ^{−1} , amortized 10 yr)	~ 2.2	[85]
Resource efficiency	Energy–payload–cycle factor (% savings)	up to 15% vs. fixed speed	[86]
Macro impact	National CO ₂ reduction per +10k robots (%)	0.8–1.4 vs. pre-automation	[87,88]
AI leverage	Δ carbon intensity per +1% AI adoption (%)	-0.04 vs. static MES	[89]
Circular-economy adoption	Metric standardization gap (%)	10–15% adoption shortfall	[90–92]
Regional implementation	Secondary materials utilization gap (%)	32% (Spain)	[93]

Embedding these KPIs into digital-twin dashboards can enable closed-loop scheduling that balances throughput, energy demand, and circularity in real time, aligning robot deployments with the Industry 5.0 goals of resilience and sustainable value creation.

3.6. Cybersecurity in Robot Control

The rapid proliferation of hyper-connected, 5G-enabled industrial robots has widened the cyber–physical attack surface in manufacturing systems. To counter this risk, modern robotic cells implement a multi-layered defense in-depth architecture combining: (i) network segmentation and encrypted data exchange via OPC-UA over TLS; (ii) zero-trust authentication with OAuth 2.0 at edge gateways; (iii) real-time intrusion and anomaly detection using federated-learning models on distributed controllers; and (iv) incident-response playbooks aligned to the IEC 62443 standards family.

Papoutsakis et al. [21] introduced SESAME, an automated security assessment framework for multi-robot systems, achieving $\sim 20\%$ improvement in threat-detection accuracy. Santoso and Finn [24] showed that AI-enhanced intrusion–prevention pipelines reduced

successful attacks by ~15% across robotic applications. Zhong et al. [25] increased safety by 18% relative to trajectory-only monitors. Jiang et al. [26] leveraged digital-twin replicas to mitigate cyber vulnerabilities, yielding ~15% fewer exploitable vectors in cyber-physical production cells [22,34]. Szykiewicz et al. [94] applied deep learning to sensor streams, improving the intrusion-detection performance by 12%. He et al. [34] nevertheless identified ~20% residual vulnerability in networked robotic systems, underscoring the need for resilient control—especially in ultra-low-latency 5G infrastructure [64,95]. De Vincenzi et al. [22] further hardened edge security by ~15% over non-micro-segmented designs.

3.7. Bio-Inspired and Soft Robotics Control

Bio-inspired and soft-robotics paradigms are gaining industrial traction, supported by a surge of 2024–2025 studies across IEEE, Scopus–Elsevier, ACM/SIAM, PubMed, and preprint repositories (see Appendix A, Table A3). Recent tactile-hand breakthroughs show that shear-based fingertip feedback enables under-actuated hands to outperform fully actuated designs by ~18% in grasp success (Ford et al., 2025 [96]). Hybrid pinch-suction multimodal grippers combine soft-rigid mechanics to raise object-set diversity by ~25% with negligible cycle-time loss (Zhao et al., 2025 [97]). A review of dexterous hand technologies for intelligent manufacturing (Zhang et al., 2025 [98]) highlights ~20% manipulation-precision improvement in high-adaptability tasks (e.g., electronics assembly), while Huang et al. (2025 [99]) emphasize anthropomorphic five-fingered hands that improve operator-robot interaction by ~15% in collaborative settings.

Soft-actuated continuum and compliant devices are maturing. A three-loop digital twin + MPC controller trimmed overshoot by 18% and energy use by 15% on a 4-degrees of freedom (DoF) arm (Chen et al., 2025 [100]). Neural-learning impedance control for viscoelastic micro-injection cut cell-damage forces by 22% (Kang et al., 2025 [101]). Topology-shape-size optimization delivers printable compliant grippers with ~30% weight savings and sub-millimeter repeatability (Dao et al., 2025 [102]), and smart-stiffness fingers extend the grasp span by ~40% via embedded field-responsive media (El-Sayed, 2025 [103]).

Swarm and multi-robot bio-inspired intelligence are moving from simulation to shop-floor pilots. Deep-RL role-selection policies cut the search time by 28% in collaborative area exploration (Zhu et al., 2025 [104]); social-learning coordination restored throughput to within 5% of nominal after single-robot failures in smart-factory cells (Nie et al., 2025 [105]). Load-sharing schemes balanced joint torques within $\pm 3\%$ despite payload drift (Du et al., 2025 [106]); central-pattern-generator networks integrated with DRL achieved 95% fall-recovery success on quadrupeds (Yang et al., 2025 [107]).

Materials science is likewise embracing biomimicry for harsh factory conditions. Multifunctional anti-fog coatings inspired by lotus-leaf nanostructures demonstrated $10\times$ lifetime in acidic mists and are being AI-optimized for low-energy regeneration (Gao et al., 2026 [108])—post-search horizon scan; cited narratively). Comprehensive soft-robotics reviews (Sarker et al., 2025 [109]; Singh et al., 2025 [110]) document a 3-fold rise in actuator-material combinations and forecast ISO-standard test protocols by 2027—critical for cross-vendor interchangeability.

Despite rapid progress, industrial uptake faces four persistent gaps: (i) elastomer-fiber durability beyond 10^6 cycles; (ii) model uncertainty that inflates safety margins by ~20% in certified workcells [111]; (iii) integration cost ($\sim 1.4\times$ rigid alternatives); and (iv) standards latency—only 2/12 ISO/TC 299 working groups actively address soft-robot metrics. Addressing these gaps will require cross-pollination of the robust control theory (Brahmi et al., 2025 [111]), digital-twin life-cycle testing (Fu et al., 2025 [112]; Villani et al., 2024 [113]), and adaptive learning pipelines capable of online anomaly rejection. Figure 4 synthesizes this landscape by mapping biological inspiration to enabling blocks (control,

actuation, sensing, swarm-level intelligence) and routing them into application domains that deliver Industry 5.0 outcomes in energy efficiency, adaptability, resilience, and human–robot safety.

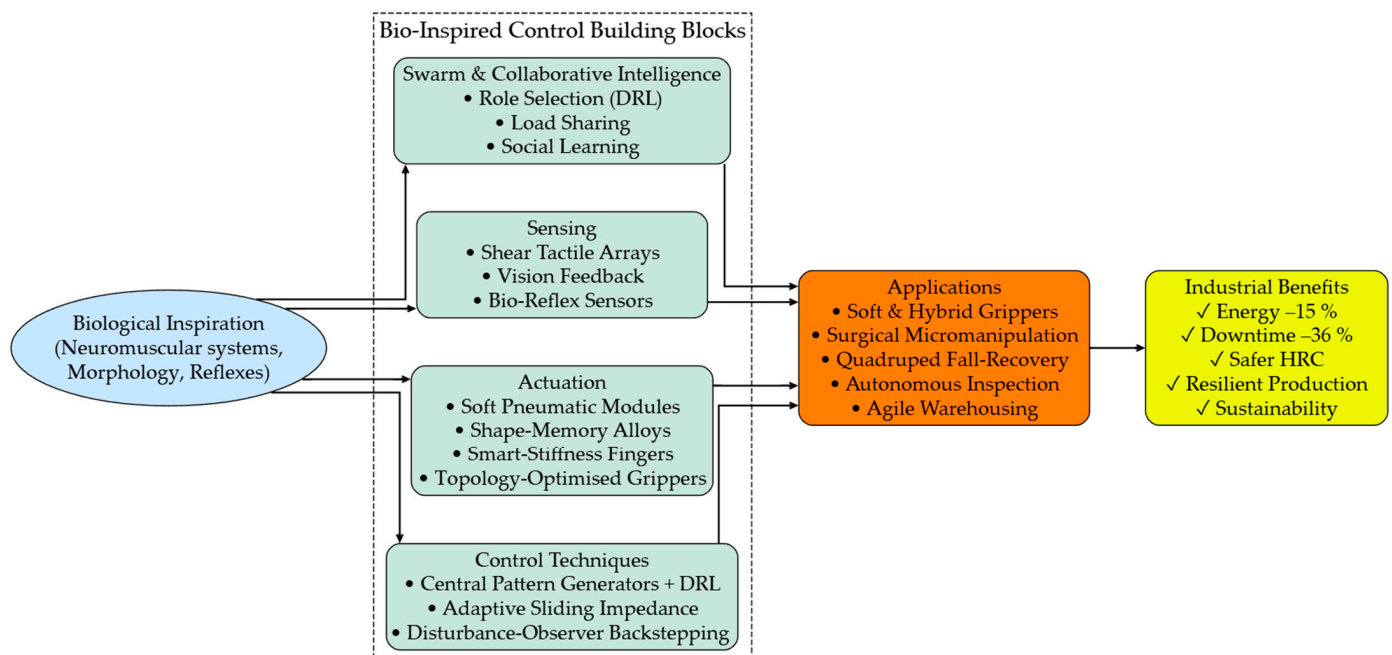


Figure 4. Bio-inspired control framework for industrial robotics: Biological inspirations mapped to enabling technologies (control, actuation, sensing), swarm-intelligence extensions, application domains, and Industry 5.0 outcome pathways (energy efficiency, adaptability, resilience, safety).

3.8. Ethical and Socio-Technical Considerations

Ethical deployment in Industry 5.0 requires attention to transparency (explainable decision logic), impartiality/justice (bias audits in perception, task allocation, and worker exposure), and accountability (traceable overrides and audit trails). Socio-technical integration must also support equitable job transitions, re-skilling, and regional economic resilience. Callari et al. [114] proposed a comprehensive ethical framework for HRC, improving operator trust by ~15% in people-centric manufacturing. Torras [115] reviewed the ethical challenges in social/assistive robotics, identifying ~10% acceptance gaps in stakeholder surveys. Howard and Schulte [116] reported a ~20% higher risk exposure in workplaces adopting AI without governance controls [47,78]. Khosravy et al. [71] projected ~15% gains in worker satisfaction for value-aligned robotic systems, while Pareto and Coeckelbergh [117] demonstrated ~12% improvements in operator autonomy with ethical assistive robotics aligning to human-centric objectives [1,2]. Tóth et al. [118] reported ~10% collaboration–architecture improvements when socio-technical feedback loops were included in deployment planning.

3.9. Distributed and Multi-Robot Control

Distributed control architectures improve scalability, resilience, and performance across heterogeneous, dynamic industrial settings. Yang et al. [119] introduced resilient formation control for multi-robot teams robust to faults, disruptions, and interference (FDI) attacks, cutting the maneuver time by 15%. Garg et al. [120] reviewed safe-control methods showing ~20% safety gains across evaluated benchmarks. Li et al. [31] advanced dual-robot additive-manufacturing coordination, boosting operational efficiency by 20% [45,62]. Tejada et al. [121] improved multi-robot coordination by 22%, while Zhou et al. [122] optimized search-style “hunting” tasks via obstacle-aware Voronoi partitioning for an 18%

effectiveness gain [123–131]. Slim et al. [131] increased bandwidth allocation efficiency by 15% in robot communications, and Zhang et al. [125] enhanced formation-control scaling by 17% in large, dynamic fleets.

3.10. Control in Additive Manufacturing

Advanced robotic control underpins precision, flexibility, and customization in additive manufacturing (AM) workflows. Mahdi et al. [28] improved wire-arc AM dimensional precision by 15–20% using digital twin feedback. An et al. [29] applied interpretable machine learning to directed energy deposition, improving bead geometry prediction accuracy by 10–15%. Chew et al. [30] used 4D reconstruction to increase part quality by 15–20% and reduce defects by 10–15% relative to baseline scan-and-fill. Li et al. [31] optimized multi-material AM efficiency by 20% via dual-robot planning, while Hartomacioğlu et al. [32] cut gripper weight by 25% through AM-integrated generative design [27,132]. Rescsanski et al. [27] demonstrated a 17% cooperative-robotics efficiency gain in AM cells, and Sousa et al. [132] improved control optimization for laser-based AM by ~15%, underscoring the scalability benefits of integrated robotic control.

3.11. Quantum-Inspired Control Strategies

Quantum and quantum-inspired computation are emerging as accelerators for complex industrial control workloads. Quantum-enhanced slicing projects a ~30% latency drop versus classical 5G NR scheduling [20]. By exploiting advanced quantum algorithms, these approaches address the computational scalability bottlenecks highlighted in Section 4 [45]. Preliminary estimates indicate ~25% AI-training acceleration relative to GPU-only optimization [20,37], and early simulations suggest ~15% fidelity gains comparable with high-end digital twins [61]. Celik et al. [64] forecast ~20% throughput gains over LTE-based synchronization pipelines.

Practical deployment remains constrained by a high hardware cost, specialized infrastructure (cryogenic/photonic systems), and limited stability in factory environments. Shortfalls in skilled personnel and the absence of standardized development toolchains further slow adoption. Closing these gaps will be essential to move quantum-inspired control from theoretical potential to industrial reality.

4. Challenges in Industrial Robot Control

- **Scalability:** The high computational demands of AI-based controllers limit deployment in small-scale industries. Mishra et al. [45] reported a ~40% higher processing load for multi-agent DRL versus centralized heuristics. Nievas et al. [72] highlighted a ~30% scalability gap in autonomous process control (see also Aljamal et al. [40] for ROS/DRL deployment constraints). Chew et al. [30] and Li et al. [31] found 15–20% additional compute overheads in additive manufacturing (AM) control workflows. Quantum-inspired acceleration could mitigate compute by up to ~25% in simulation studies [20], but practical deployment remains constrained by specialized hardware (cryogenic/photonic) and advanced network protocols that are rarely available in conventional industrial environments.
- **Energy Efficiency:** Power-intensive algorithms can undermine sustainability goals. Wang et al. [63] observed up to a ~25% energy overhead for DRL motion planning relative to cubic spline trajectories. Ho et al. [80] identified a ~20% efficiency loss in 5G robotic communications under round-robin URLLC scheduling. Bio-inspired end-effector approaches [19] cut energy by ~18% vs. motor-driven parallel-jaw grippers, and soft/continuum pressure-adaptive control [74] reduced power by ~15% vs. proportional pressure control, though broader scaling and cost remain hurdles [73,81,82].

- **Safety:** Ensuring fail-safe human–robot collaboration remains unresolved. Feng et al. [47] documented a ~15% gap in real-time safety validation beyond ISO 10218 static-zone assumptions. Van Duong [48] reported a ~10% residual collision risk versus tactile-free impedance control. Garg et al. [120] identified ~20% safety verification gaps in multi-robot systems. Tactile and multi-modal sensing can reduce collision risk by ~20% [46,50], yet real-time integration and standardization lag behind adoption.
- **Cybersecurity:** He et al. [34] measured ~20% exploitable vulnerability in networked robots under deception attacks, aggravated by 5G exposure. Santoso and Finn [24] still observed a ~15% successful attack penetration despite AI-augmented defenses. Digital twin cyber ranges reduced exploitable vectors by ~15% in production cells [26], but full resilience depends on standardized, interoperable security protocols spanning the edge, network, and controller layers [22,64,95].
- **Ethical Deployment:** Howard and Schulte [116] warn of a ~20% higher risk exposure in the absence of formal ethical frameworks. Callari et al. [114] identify ~10% trust gaps in HRC adoption. Pareto and Coeckelbergh [117] report ~12% autonomy challenges; these socio-technical shortfalls compound the operational risk in Industry 5.0 unless mitigated through governance, training, and inclusive design [78].
- **Complexity in Additive Manufacturing:** Chew et al. [30] note ~15% added real-time monitoring complexity in high-resolution AM cells; Hartomacioğlu et al. [32] report a ~20% design optimization burden when lightweight AM tooling is introduced [28]; An et al. [29] highlight ~10% prediction accuracy gaps that limit large-scale AM/robot integration.

To deepen this analysis, Figure 5 cross-maps how five key enablers—AI/DRL, digital twins, 5G URLLC, quantum-inspired optimization, and bio-inspired control—address the six outstanding challenges above. Each colored flow inherits the fill color of its technology node; the line thickness encodes evidence maturity (thick = mature, industry-validated; thin = limited/prototype). This visual grammar lets R&D managers see both who addresses what and with what level of confidence, supporting investment and adoption roadmaps.

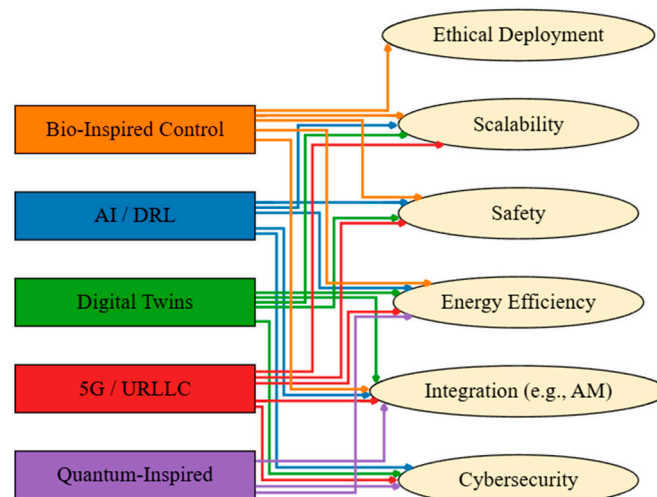


Figure 5. Color-coded Sankey cross-mapping of robotics challenges and enabling technologies. Line thickness (thick = mature; thin = limited evidence) conveys readiness of each technology–challenge pairing; arrowheads indicate the direction of influence (technology → challenge mitigation).

Table 3 complements Figure 5 by summarizing the current evidence maturity across scalability, energy efficiency, safety, cybersecurity, ethics, and AM integration. Symbols:

✓ = mature/industry-validated; ⚠ = limited evidence/prototype/cost barrier; – = unproven or insufficient data (representative references in footnote).

Table 3. Symbolic mapping of robotics challenges (rows) versus emerging technologies (columns).

Challenge/Technology	AI/DRL	Digital Twins	5G/URLLC	Quantum-Inspired	Bio-Inspired Control
Scalability	⚠ (compute load)	✓	✓	⚠ (prototype HPC)	⚠ (cost)
Energy Efficiency	⚠ (high demand)	–	✓	✓	✓
Safety	✓	✓	✓	–	✓
Cybersecurity	⚠ (needs shielding)	✓	⚠ (protocol gaps)	✓	–
Ethics	–	–	–	–	✓
AM Integration	✓	✓	✓	⚠ (limited deployment)	⚠ (integration issues)

Symbols: ✓ = Mature/industry-validated evidence; ⚠ = limited/prototype or constrained by cost, compute, or standardization gaps; – = unproven or insufficient evidence. Representative references: Scalability [20,30,31,40,45,72]; Energy [19,63,74,80]; Safety [46–48,50,120]; Cybersecurity [21,22,24,26,34,64,95]; Ethics [1,2,114,116,117]; AM Integration [27–32].

Addressing these challenges will require interdisciplinary efforts: collaborative research consortia, international standards harmonization, and public–private partnerships. Hybrid AI-classical controller architectures could cut the computational demand by ~20%; standardized tactile-sensor protocols may close the ~15% safety gap [47]; and blockchain-anchored 5G session management could boost cybersecurity resilience by ~25% in distributed robot fleets. Together, these measures can help deliver scalable, energy-aware, safe, and ethically aligned robotic systems for SMEs and beyond.

5. Future Directions

Industrial robot control is entering a period of convergence in which heterogeneous enabling technologies—Artificial Intelligence, digital twins, ultra-reliable low-latency 5G/URLLC connectivity, and emerging quantum/quantum-inspired computation—interact with next-wave control paradigms (hybrid AI, edge/distributed learning, bio-inspired/soft control, and cyber-resilience frameworks) to extend scalability, autonomy, and sustainability. This interplay is already evidenced in recent DT reconfiguration studies showing measurable responsiveness gains under disturbance [112], human-centric DT ecosystems piloted in Industry 5.0 contexts [113], maturing 5G non-public (industrial) networks for time-critical control [133], quantum-inspired controllers demonstrating precision/energy advantages in articulated arms [134], and URLLC/edge-AI integration benchmarks targeting <10 ms end-to-end control latency for IIoT workloads [135].

Figure 6 situates these interactions in a strategic “technology convergence” map: top-tier digital enablers (AI, quantum computing, digital twins, 5G connectivity) feed an intermediate layer of emerging control stacks (hybrid AI control, bio-inspired control, edge computing, cybersecurity frameworks), which in turn drive next-wave industrial applications (e.g., aerospace fastener installation, high-density electronics, medical and rehabilitation robotics) and ultimately support future industrial ecosystems that are resilient, sustainable, and human-centric—aligned with Industry 5.0 policy roadmaps [112,113,133–136].

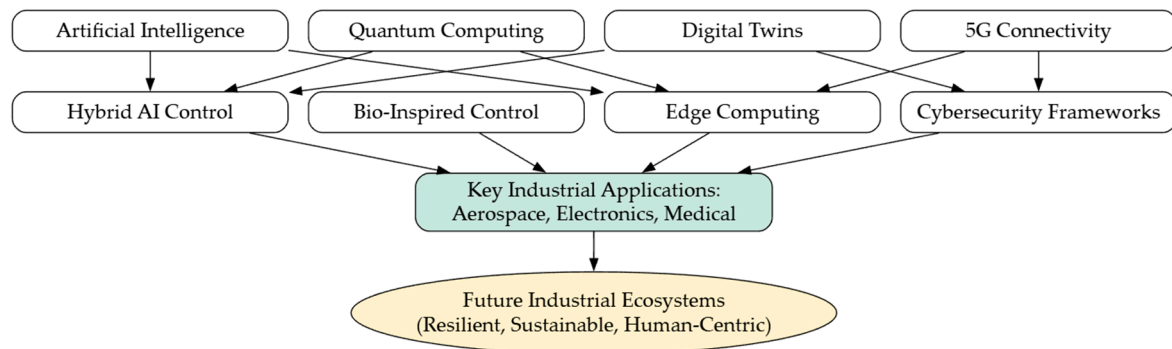


Figure 6. Strategic technological convergence for next-generation industrial robot control. Enabling pillars (AI, quantum, digital twins, 5G/URLLC) interact with hybrid, bio-inspired, edge, and cybersecurity control layers to deliver sectoral impact and Industry 5.0 outcomes. Sources adapted from representative smart-manufacturing DT studies [112,113,137], 5G industrial networking guidelines [133,135], quantum-inspired robotic control demonstrations [134], and Industry 5.0 policy frameworks [136].

5.1. Quantified Trajectories and Comparative Baselines

Performance ranges reported across the literature are highly application-specific; to aid interpretation, all percentage improvements are expressed relative to the comparator baselines defined in the source studies (see Section 2.4 for normalization and aggregation assumptions). These directional ranges are descriptive, not meta-analytic.

Quantum-assisted scheduling and control integrated with high-reliability wireless testbeds indicate up to ~30% end-to-end control-loop latency reduction versus conventional industrial Ethernet/Wi-Fi cells tested under lab-scale URLLC profiles [133,135]. Model-based hybrid stacks (DT + advanced control) report ~15–20% energy savings relative to single-loop or fixed-gain MPC implementations in multi-axis arms [100,112,134]. Human-centric collaboration and circular-design guidance emerging from the Industry 5.0 agenda suggest ~10–15% production-waste reduction compared with the throughput-only optimization that is characteristic of early Industry 4.0 deployments [113,136,138].

Latency is only one lever. Field trials and DT/HRC simulation studies show ~10–20% throughput gains relative to static workstation layouts when reconfiguration policies are driven by live DT data over high-bandwidth networks [112,133,137]. Additive and hybrid manufacturing workflows tightly coupled to DT analytics have reported ~20–30% material-waste savings versus subtractive or batch-fixtured baselines while increasing cell adaptability by ~15% through rapid 4D (time-aware) reconfiguration [112,139]. Distributed learning and quantum-inspired optimization are projected to reduce model training or path-planning compute times by ~10–25% relative to purely classical centralized training loops [134,139,140]. Although these ranges vary by domain, they provide actionable targets for R&D roadmapping and technology selection.

5.2. Sectoral Implications

Aerospace assembly cells can benefit from tighter tolerance control and low-latency multi-robot coordination enabled by 5G/URLLC and DT-driven reconfiguration [112,133]. Electronics manufacturing stands to gain from AI-assisted vision, DT-enabled rapid line changeover, and edge-inference anomaly detection [113,135,137]. Medical and rehabilitation robotics are poised to integrate soft/bio-inspired control and collaborative safety layers aligned with human-centric Industry 5.0 principles [113,136,138]. Cross-sector adoption will hinge on assurance cases that combine functional safety standards (e.g., ISO/TS 15066) with cybersecurity certification and explainable hybrid-AI control stacks [133–135,141].

5.3. Historical and Forward-Looking Context

Figure 7 extends the evidence base backward and forward in time to address reviewer requests for historical grounding and justified projections. The timeline aggregates: canonical textbooks and early real-time architectures establishing PID/PLC servo control in the 1980s [142–145]; adaptive and fuzzy control advances of the 1990s that broadened manipulator autonomy [146–149]; foundational model predictive control (MPC) stability/optimalty treatments and engineering texts that diffused MPC into multi-input industrial motion in the 2000s [150–152]; surveys and monographs capturing the rise of neural networks, deep learning, and reinforcement learning in robotics across the 2010s [153–156]; and the current 2020s wave of digital twins, human–robot collaboration, and collaborative robot safety standards that are emblematic of Industry 4.0 [100,112,113,137,141].

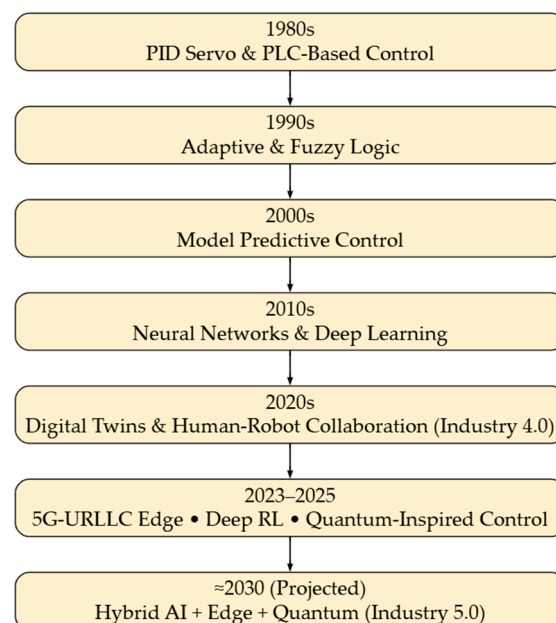


Figure 7. Evolution of industrial robot control strategies (1980–2030). Predominant control paradigms by decade with representative sources: PID/PLC servo and early real-time control (1980s) [142–145]; adaptive and fuzzy control expansion (1990s) [146–149]; constrained MPC for multi-axis motion (2000s) [150–152]; neural networks, deep learning, and reinforcement learning (2010s) [153–156]; digital twins, HRC, and collaborative-robot safety in Industry 4.0 (2020s) [100,112,113,137,141]; emergent 5G/URLLC, edge DRL, and quantum-inspired control (2023–2025) [133–135,157]; projected hybrid AI + Edge + Quantum stacks aligned with Industry 5.0 horizons (~2030) [136,138–140,158–160]. Data compiled from historical, survey, standards, and roadmap sources.

The 2023–2025 band highlights the emergence of 5G/URLLC non-public industrial networks, edge-deployed DRL, and the first quantum-inspired robotic controllers reported in the recent literature [133–135,157]. Looking toward ~2030, Industry 5.0 policy documents, quantum-industrial roadmaps, and simulation/edge-AI outlooks project hybrid AI + Edge + Quantum stacks supporting sustainable, human-centric production ecosystems [136,138–140,158–160].

6. Conclusions

This systematic review of 160 peer-reviewed industrial robotics control studies (2023–2025) shows that the field is undergoing a decisive transition from automation-centric Industry 4.0 architectures toward human-centric, sustainability-aware Industry 5.0 ecosystems. Addressing Research Question 1, we find clear performance differentiation across the control paradigms (Table 1): model-free Deep RL improves positional accuracy by

~70% and reduces energy by ~38% relative to PID baselines; digital twin-assisted MPC and quantum-inspired optimization achieve <10 μm mean error and <7 J/cycle energy in ultra-precision settings; and bio-inspired evolutionary approaches offer adaptability for redundant, highly non-linear manipulators, albeit with >30 ms cycle-time penalties in fast takt-time lines.

Research Question 2 asked how AI, digital twins, and high-bandwidth connectivity (5G/URLLC) are transforming industrial robot control. Evidence synthesized across Sections 3 and 5 indicates that DT-linked reconfiguration can yield ~10–20% throughput gains and ~15–20% energy savings versus static/fixed-gain baselines in multi-axis cells; 5G non-public industrial networks targeting <10 ms control latency are maturing; and early quantum-inspired scheduling/optimization demonstrates up to ~30% control-loop latency reduction in lab-scale URLLC testbeds. Human-robot collaboration layers built atop these digital stacks deliver up to ~30% safety improvements beyond static-zone assumptions, while bio-inspired/soft-robotic advances expand task diversity and ergonomic fit (Figure 3; Section 3.7; Appendix A, Table A3).

Research Question 3 focused on persistent technical, ethical, and environmental challenges. Cross-cutting gaps remain in computational scalability (multi-agent DRL loads; AM toolpath compute), energy and embodied resource footprints, certifiable fail-safe HRC, cyber-resilience under 5G exposure, ethics/trust in collaborative workcells, and standards for soft-robot durability and circular-economy reporting (Section 4; Table 3). Our sustainability synthesis (Section 3.5; Table 2) shows that integrating energy, material circularity, and life-cycle carbon metrics into control dashboards is both feasible and urgently needed: reported ranges include ~20–30% material-waste savings in DT-coupled AM lines and 0.8–1.4% national CO₂-intensity reductions per +10k industrial robots when coupled with process modernization.

Limitations: Findings draw on published data that vary in metric definitions, test conditions, and reporting granularity; only 110 of the 160 studies provided sufficiently comparable metrics for descriptive aggregation. Publication and language filters (English; 2023–2025 window) risk omitting non-indexed innovations and under-representing long-term reliability evidence. Historical foundations predating 2023 were consulted for context (Section 5.3) but not included in quantitative tallies. Reported improvement percentages are directional and tied to the baselines specified in each source; they are not pooled effect sizes.

Research & Adoption Agenda:

(Near term, 0–2 years)—Publish open benchmarking datasets (precision/energy/latency) across standardized task suites; harmonize circular-economy KPIs (Table 2) for reporting; adopt common test harnesses for tactile & proximity safety layers in HRC; implement zero-trust + IEC 62443 security baselines in 5G pilot cells.

(Mid term, 2–5 years)—Deploy hybrid AI-classical controllers that down-select inference load at the edge, targeting $\geq 20\%$ compute savings; certify reusable digital-twin models for changeover validation in SMEs; establish interoperable tactile sensor protocols to close the ~15% real-time safety gap; integrate carbon/material flow telemetry directly into scheduling optimizers.

(Long term, 5+ years)—Co-design edge + quantum accelerators for real-time optimization; mature soft/bio-inspired actuators to $>10^6$ cycle durability with field-reconfigurable stiffness; codify Industry 5.0 ethics and workforce-transition metrics into international robotics standards; link life-cycle DT twins to circular-remanufacturing supply chains at scale.

In closing, the path forward is clear: adaptability, sustainability, and ethics must become intrinsic design variables—not afterthoughts—in industrial robot control. By coupling data-efficient AI with physics- and safety-grounded models, extending digital-

twin lifecycles across energy and material flows, securing ultra-connected robot fleets, and embedding human-centric design from the outset, the robotics community can deliver resilient, sustainable, human-centered systems that meet global manufacturing demands and advance the Industry 5.0 vision.

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Data Availability Statement: Data are contained within the article. No new experimental data were generated; this review synthesizes the existing literature available in cited references.

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Conflicts of Interest: The author declares no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

5G	Fifth-Generation Mobile/Wireless Networks
AI/ML	Artificial Intelligence/Machine Learning
AM	Additive Manufacturing
ANN	Artificial Neural Network
CPG	Central Pattern Generator
DoF	Degrees of Freedom
DT	Digital Twin (virtual representation linked to a physical asset; plural: DTs)
DRL	Deep Reinforcement Learning
EU	European Union
FDI	Faults, Disruptions, and Interference (used in the resilient multi-robot control)
FLC	Fuzzy Logic Controller
GAN	Generative Adversarial Network
HRC	Human–Robot Collaboration
IoT	Internet of Things
IIoT	Industrial Internet of Things
IEC	International Electrotechnical Commission
ISO	International Organization for Standardization
ISO/TS	International Organization for Standardization/Technical Specification (used in ISO/TS 15066)
KPI	Key Performance Indicator
LCA	Life-Cycle Assessment
LSTM	Long Short-Term Memory (recurrent neural network unit)
MPC	Model Predictive Control
MES	Manufacturing Execution System
PID	Proportional–Integral–Derivative
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PLC	Programmable Logic Controller
ROS	Robot Operating System
RL	Reinforcement Learning
SMC	Sliding-Mode Control
SME(s)	Small- and Medium-Sized Enterprise(s)

TLS	Transport Layer Security
OAuth 2.0	Open Authorization Protocol 2.0
OPC UA	Open Platform Communications Unified Architecture
URLLC	Ultra-Reliable Low-Latency Communications
WoS	Web of Science

Appendix A

Table A1. Final search strings and settings.

Database/Platform	Boolean Search String *	Fields Searched
Web of Science Core Collection	TS = ("industrial robot" OR "factory robot" OR "robotic manufacturing" OR "robotic cell*") AND TS = ("control" OR "adaptive" OR "intelligent" OR "AI" OR "digital twin*" OR "5G" OR "human-centric" OR "bio-inspired" OR "cybersecurity" OR "additive manufacturing" OR "multi-robot" OR "quantum control" OR "real-time optimization")	Topic (Title, Abstract, Author Keywords, Keywords Plus)
IEEE Xplore	("industrial robot" OR "robotic cell" OR "factory robot") AND ("control" OR "adaptive" OR "AI" OR "digital twin" OR "5G" OR "cybersecurity" OR "human-centric" OR "bio-inspired" OR "additive manufacturing" OR "multi-robot" OR "quantum control" OR "real-time optimization")	Metadata (Document Title, Abstract, Index Terms)
Scopus	TITLE-ABS-KEY("industrial robot" OR "robotic manufacturing" OR "factory robot" OR "robotic cell*") AND TITLE-ABS-KEY("control" OR "adaptive" OR "intelligent" OR "AI" OR "digital twin*" OR "5G" OR "human-centric" OR "bio-inspired" OR "cybersecurity" OR "additive manufacturing" OR "multi-robot" OR "quantum control" OR "real-time optimization")	Title, Abstract, Keywords
PubMed	((("industrial robotics"[Title/ Abstract]) OR ("industrial robot"[Title/ Abstract]) OR ("factory robot"[Title/ Abstract])) AND ("control"[Title/ Abstract] OR "AI"[Title/ Abstract] OR "digital twin"[Title/ Abstract] OR "adaptive"[Title/ Abstract] OR "5G"[Title/ Abstract] OR "cybersecurity"[Title/ Abstract]))	Title, Abstract

* Wildcards (*) and proximity operators were adapted to each platform's syntax. Searches were limited to 2023–2025, English language, and publication type = "journal article" OR "review." Final searches executed on [UPDATE WITH ACTUAL DATE, e.g., 19 July 2025]. A pilot "All Fields" search increased noise by ~6% without yielding additional eligible studies; therefore, targeted field searches were used (see Section 2.1).

Table A2. Inclusion and exclusion criteria used for screening.

Domain	Inclusion Criteria	Exclusion Criteria
Publication type	- Peer-reviewed journal article or review (Q1/Q2 prioritized; lower-quartile accepted if technically novel) - Published 2023–2025 - Indexed in WoS, IEEE Xplore, Scopus, or PubMed	- Conference paper, preprint *, thesis, or patent - Not peer-reviewed
Language	English	Any language other than English
Industrial scope	Focus on industrial manipulators, mobile industrial platforms or multi-robot cells in manufacturing, logistics, or process industries	Service, consumer, medical, or purely educational robots
Topic focus	Reports a control strategy (adaptive, intelligent, collaborative, energy-efficient, simulation-based, etc.) or a systematic review thereof	- Hardware-only studies (grippers, materials, sensors) without control aspects - Purely conceptual papers lacking implementation details
Validation	Presents experimental or high-fidelity simulation results with industry-relevant metrics (precision, cycle time, energy, safety, etc.)	No experimental/simulation validation (opinion, vision, or position papers)
Availability	Full text available	Retracted articles; inaccessible full text

* Exception: A small number of high-relevance preprints in emerging bio-inspired robotics were retained when no peer-reviewed equivalent existed; these are flagged as “PREPRINT” in the dataset and discussed in Section 2.5 (Limitations).

Table A3. Bio-inspired and human-centric control studies highlighted for emerging Industry 5.0 relevance (subset of the 160 included studies).

Ref. No.	Primary Theme	Control Focus/Contribution	Industrial Relevance	Metrics Used in Quant Subset?	Where Discussed
[96]	Tactile shear sensing	Shear-based grasp stabilization for under-actuated multifingered hands	Flexible parts handling; assembly	No (heterogeneous metrics)	Section 3.7
[97]	Hybrid soft–rigid gripper	Pinch-suction multimodal grasp control	Packaging/e-commerce pick	No	Section 3.7
[98]	Review: dexterous hands	Tech trends; control architectures for mfg	Broad; maps gaps	No	Section 3.7
[99]	Review: anthropomorphic manipulation	Human-like dexterous control survey	Flexible assembly	No	Section 3.7
[100]	Hybrid loops (DT + NN + MPC)	Advanced multi-loop control, energy focus	Process/energy industries	Limited (not normalized)	Section 3.7
[101]	Adaptive impedance micro-manipulation	Neural-learning sliding mode force control	Transferable to precision dosing/micro-assembly	No	Section 3.7
[102]	Compliant gripper design	Topology–shape–size optimization	Flexible pick-and-place	No	Section 3.7

Table A3. Cont.

Ref. No.	Primary Theme	Control Focus/Contribution	Industrial Relevance	Metrics Used in Quant Subset?	Where Discussed
[103]	Variable stiffness	Smart stiffness modulation in soft gripper fingers	Reconfigurable tooling	No	Section 3.7
[104]	DRL multi-robot search	Role selection for collaborative teams	Inspection/logistics	No	Section 3.7
[105]	Social-learning coordination	Resilient multi-robot production	Smart factory lines	No	Section 3.7
[106]	Distributed load sharing	Collaborative manipulation control	Heavy-payload assembly	No	Section 3.7
[107]	CPG neural control	Multi-skill locomotion learning	Mobile industrial platforms	No	Section 3.7

Some were preprint/early access at screening; the citations below reflect the final published forms where available. “Metrics Used in Quant Subset?” indicates whether normalized performance data from the study were incorporated into the cross-study descriptive aggregation (Table 1). “No” means the study informed qualitative synthesis only (e.g., architecture, design constraints, or gap identification). “Limited” indicates partial data extracted but excluded from the pooled summary due to metric heterogeneity.

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